Run report: rho reference and replay for the plain-S_5 C_4 row, 2026-10-21

Descriptive record of the validation runs behind `../../README.md` (“Validation”). Observations only.

- **Binary:** `examples/gaudry_quartic_c4_options.rs`, a fresh release build of `ca88b3917` in a new target directory (rustc 1.93.1, Homebrew).
- **Host:** Apple M4 Pro, 14 logical cores, macOS arm64, `RAYON_NUM_THREADS=4`, **no isolation tier**. The host load average was about 240–300 from other sessions’ jobs. Counted `F_p` multiplications don’t depend on contention. Wall times are not evidence.
- **Instances:** `p = 269`, seed 1, the curve and residual stream of §11.17.

run (UTC start)	what	requested / completed	failures	record
2026-10-21 05:01Z	`--rho 20`: #E from point orders in the Hasse interval, then 20 one-target negation-map rho solves in the prime subgroup	20 / 20	none	`validation/rho_reference.json`
2026-10-21 05:01Z – 10:00Z	`validation/replay.sh`: replay of the four recorded S_5/generic runs	71 / 71 solves	none	`validation/replay/*.json, .log`
2026-10-26 01:15Z	`validation/replay.sh`: replay of the rho runs, then the certificate	20 / 20	none	`validation/replay/rho_reference.json`, `validation/replay-certificate.json`

Also recorded: `validation/curve_info.json` from `examples/gaudry_quartic_curve_info.rs`, with `j(E)` and its subfield tests.

Recorded values.

- **Group.** `#E(F_{269⁴}) = 5,236,052,031 = 189 · 27,703,979`, with ℓ = 27,703,979 prime (24.72 bits). Setup, outside the runs: 118,498 `F_p` multiplications.
- **Rho, per run.** `F_p` multiplications 331,962 – 1,216,863, mean 773,556.2 (n = 20). Steps 717 – 9,570. Look-aheads 13 – 174. Restarts 0 – 3. S (group operations / $\sqrt{\ell}$, setup multipliers included) 0.60 – 2.34, mean 1.47. All 20 logarithms were recovered and verified.
- **Curve.** `j(E)` is in neither `F_p` nor `F_{p²}`, and `a, b ≠ 0`.
- **Replay.** 91 / 91 solves identical: every counted phase, every decomposition and oracle set, and every rho run’s counts and verified logarithm. Certificate status `verified`.
- **Validator.** 17 / 17 headline values recomputed, status `pass`.
- **§5a band** (validator recompute, `F_p` multiplications): rho $7.736 \cdot 10^5$, $1.025 \cdot 10^6$ and $3.289 \cdot 10^6$ at w = 0, 0.1 and 1. The IC lower bound is $8.690 \cdot 10^{10}$ at every w.

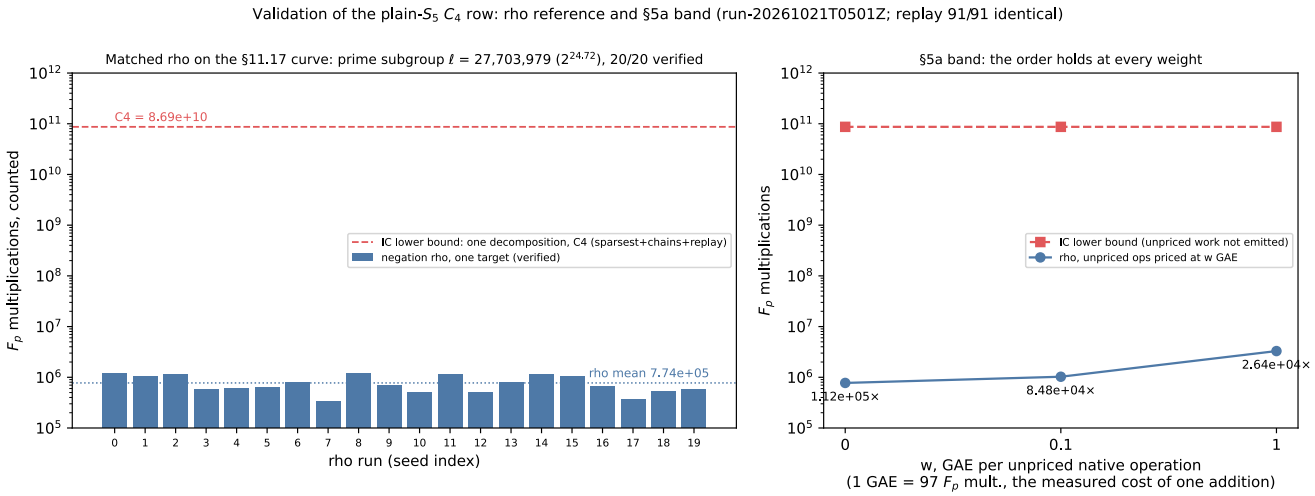


Figure 1: rho reference and band

Figure: the 20 rho runs’ counted `F_p` multiplications against the IC lower bound (left); the §5a band (right). Run run-20261021T0501Z.

Graphs checked and left unchanged: `figures/c4_options.svg` (the C_4 arms). It has no rho data and its y-axis starts at 10^{10}, so the rho values (~10^6) belong in the new figure instead.